

Md Imran Khan

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PROFESSIONAL SUMMARY

Metrology Engineer with Ph.D. training in optics and nearly 4 years of semiconductor high-volume manufacturing experience in CD-SEM metrology at Intel. Proven expertise in metrology fleet management, gauge capability (GR&R), SPC-driven process optimization, and Python-based data automation for high-volume manufacturing. Expertise in optical physics, image analysis, and data-driven root cause investigations.

TECHNICAL SKILLS

Metrology & Inline QC: CD-SEM fleet management, gauge capability (GR&R / Cp / Cpk), SPC, tool-to-tool matching, measurement uncertainty, inline metrology deployment & commissioning

Characterization: SEM, CD-SEM, Optical Microscopy, TRPL, Image Quality Analysis (CNR, Sharpness, FFT, GLCM)

Optics & Sensors: Electron-optical calibration, optical sensor design & alignment, Optical simulation (MFS, Foldy-Lax), TRPL system design

Programming & Data: Python (PyTorch, OpenCV, scikit-image, NumPy, pandas, Matplotlib), SQL/MySQL, Git, MATLAB, LabVIEW, JMP, JSL (JMP Scripting Language)

ML & Image Processing: Anomaly detection, semantic segmentation (U-Net), SSIM/PSNR/FFT image quality metrics, multivariate statistical pipelines

Fabrication: Clean room, sputtering, photolithography, nano-fabrication, DLS

EXPERIENCE

Metrology Process Engineer – CD-SEM

November 2021 – August 2025

Intel Corporation | Hillsboro, OR

- Managed fleet-wide gauge capability and measurement stability for 8 CD-SEM systems in high-volume semiconductor manufacturing; applied SPC, automated calibration logic, and gage R&R studies to sustain Cp/Cpk targets and minimize measurement drift across process nodes.
- Designed and deployed a Python-based anomaly detection system monitoring high-dimensional sensor data (beam parameters, sub-system trends), transitioning team from reactive troubleshooting to proactive real-time optimization — directly analogous to inline metrology health monitoring for battery electrode processing lines.
- Automated multivariate statistical pipelines for tool-to-tool matching and measurement precision evaluation on DCCD processes; reduced data processing time by 80% (5 hrs to 1 hr) while improving fleet-wide consistency reporting accuracy.
- Defined measurement sampling strategies and gauge capability specifications to qualify new process layers and product lines following tool installation or process integration changes.
- Performed structured root cause investigations on electron-optical drift and image acquisition anomalies; coordinated across hardware, software, and process integration teams to resolve complex metrology-process interactions.
- Led metrology qualification activities for new CD-SEM installations, including system calibration, GRR validation, tool matching, and production release readiness.
- Developed Python workflows integrating FCCD datasets into long-term SPC systems, enabling data-driven process optimization and visualization dashboards for cross-functional review.
- Authored metrology documentation and training materials; trained technicians on tool maintenance, calibration procedures, and SPC interpretation to ensure production readiness across shifts.
- Collaborated with cross-functional team for SEM metrology stability, coordinating with hardware and software integration teams to troubleshoot complex electron-optical drift and optimize image acquisition.

RESEARCH EXPERIENCE

Graduate Research Scholar – Optics & Computational Modeling August 2016 – October 2021

University of California, Merced | Merced, CA

- Designed and commissioned a high-stability TRPL optical measurement system optimized for weak emitters; defined system specifications, aligned optical components, and validated measurement performance against simulation — applicable to novel inline optical sensor development.
- Developed a hybrid full-wave modeling framework (MFS and Foldy-Lax theory) to simulate 3D nanoplasmonic scattering and broadband optical response across UV–IR wavelengths; demonstrated strong agreement between simulation and experimental measurements.
- Nano-fabricated core-shell meta-structures and correlated experimental scattering data with computational models, bridging hands-on fabrication with quantitative metrology validation.

TECHNICAL PROJECTS

SEM Semantic Segmentation for Microstructure Analysis | *PyTorch, Python, PIL*

- Built a multi-class semantic segmentation pipeline (U-Net) for SEM microstructure images to segment 5 material phases; implemented class-weighted CE + Dice loss and per-class IoU/Dice evaluation — enabling automated microstructure quantification applicable to battery electrode quality control.

SEM Image Quality Analysis Toolkit | *Python, OpenCV, scikit-image*

- Developed an imaging sensor characterization toolkit implementing full-reference (SSIM, PSNR) and no-reference (Laplacian sharpness, FFT, variance-based focus) metrics for automated image quality monitoring — directly applicable to inline camera-based metrology systems.

INTERNSHIP

Research Project Assistant | Universal Instruments Corporation, Kirkwood, NY | 2013–2014

- Investigated the effect of Sn surface finish on melting temperature, microstructure, and mechanical behavior of 92.5Pb-5Sn-2.5Ag solder alloys using DSC and SEM characterization.

EDUCATION

Ph.D., Physics – Optics & Computational Modeling

University of California, Merced

October 2021

M.S., Physics

Binghamton University (SUNY)

May 2016

SELECTED PUBLICATIONS

Khan, M.I., et al., "Scattering by nanoplasmonic mesoscale assemblies," *JOSA A* 42, 1244–1253 (2025).

Khan, M.I., et al., "Modeling broadband cloaking using 3D nano-assembled plasmonic meta-structures," *Optics Express* (2020).

Schoeller, H., Anselm, M., Khan, I., & Cotts, E., "Effect of Sn Component Surface Finish on 92.5Pb-5Sn-2.5Ag," *HITEC Conf. Proc.*, Vol. 2014, pp. 364–371.

AWARDS & AFFILIATIONS

- NSF-CREST Graduate Scholar, Center for Cellular and Biomolecular Machines (CCBM), UC Merced
- Graduate Scholar Mentor — mentored undergraduate researchers on experimental methods and graduate school preparation